

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B.Tech (EE) Examination

December 2010

EE 202 Electrical Power Generation

Date: 27.11.2010 ,Saturday

Time:10.30 a.m to 1.30 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Figures to the right indicate *full* marks.
3. Make suitable assumptions and draw neat figures wherever required
4. Attempt both section Separately

FOR REFERENCE



SECTION- I

- Q1 (a) Explain steam water cycle of thermal power plant with sketch. 06
- (b) Which are major auxiliaries of steam power plant? Explain their functioning? 05

- Q2 (a) Give the general layout of a hydro-electric power plant. Clearly explain the working of a hydro-electric plant. 06
- (b) Discuss advantage & disadvantage of thermal power plant. 03
- (c) Discuss merits & demerits of Hydro power plant. 03

OR

- Q2 (a) What is a fuel cell? Describe the principle of working of a fuel cell with reference to H₂ – O₂ cell. 06
- (b) Classify the types of the energy sources available on the earth. 03
- (c) What is meant by pumped storage plant? Explain its importance. 03

- Q3 (a) Explain the functioning of closed cycle Gas turbine power plant with the help of neat sketch? 06
- (b) Compare the diesel power plant and gas power plant of the same capacity 03
- (c) Explain fuel cycle of thermal power plant. 03

OR

- Q3 (a) Give principle of ocean thermal energy conversion system, explain closed cycle OTEC pant. 06
- (b) Explain how with net sketch the use of tidal power is made to generate electricity? 06



SECTION – II

- Q4 (a) Explain with neat sketch the working of a nuclear power station. 06
- (b) Discuss the advantages and disadvantages of nuclear power plant compared to steam power plants. 05

- Q5 (a) Draw schematic diagram of an MHD power generating system having heat recovery steam generator Explain function of system. 06
- (b) Explain parabolic trough solar collector type solar thermal power plant with net sketch. 06

OR

- Q5 (a) Classification of wind energy conversion system & discuss advantage, disadvantage with respect to WECS. 06
- (b) Discuss site selection of wind farm. 03
- (c) Enlist different types solar collectors, Explain any one with net sketch. 03

- Q6 (a) What is solar pond? Explain solar pond electric power plant with net sketch. 06
- (b) Explain binary cycle liquid dominated geothermal power plant. 06

OR

- Q6 (a) Draw schematic arrangement of Diesel Power station and explain each system. 06
- (b) Write short note : 06
1. Merits and demerits of diesel power plants.
 2. Solar pond

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